LOLCODE Markdown Language Context Free Grammar

The following is one possible context-free (BNF) grammar for the LOLCODE Markdown language described in Project 1. The grammar is written in plain BNF and is structured for implementation using a hand-written recursive-descent parser.

Terminals

```
HAI        ::= "#HAI"
KBYE       ::= "#KBYE"
OBTW       ::= "#OBTW"
TLDR       ::= "#TLDR"
MAEK       ::= "#MAEK"
OIC        ::= "#OIC"
GIMMEH     ::= "#GIMMEH"
MKAY       ::= "#MKAY"
HEAD       ::= "HEAD"
TITLE      ::= "TITLE"
PARAGRAF   ::= "PARAGRAF"
BOLD       ::= "BOLD"
LIST       ::= "LIST"
ITEM       ::= "ITEM"
NEWLINE    ::= "#NEWLINE"
LINX       ::= "LINX"
IHAZ       ::= "#IHAZ"
ITIZ       ::= "#ITIZ"
VAR        ::= "VAR"
VARNAME    ::= any single word (A-Z, a-z, no spaces)
VARVALUE   ::= allowed text characters
TEXT       ::= plain text (letters, digits, punctuation, spaces, tabs, newlines)
ADDRESS    ::= text without spaces
```

Notes on terminals

- All annotations are case-insensitive.
- The only allowed plain-text characters are those specified in Project 1.
- The "#" character is reserved for LOLCODE Markdown annotations and is not part of ordinary TEXT.

Syntax / Production Rules (Non-terminals)

```
<lolcode> ::= HAI <comments> <head> <optional-variable-define> <body> KBYE

<comments> ::= <comment> <comments>
             | ε

<comment> ::= OBTW TEXT TLDR

<head> ::= MAEK HEAD <title> MKAY
         | ε

<title> ::= GIMMEH TITLE TEXT OIC

<body> ::= <inner-body> <body>
          | ε

<inner-body> ::= <paragraph>
               | <comment>
               | <bold>
               | <list>
               | <link>
               | <newline>
               | <variable-use>
               | TEXT

<paragraph> ::= MAEK PARAGRAF <optional-variable-define> <inner-paragraph> MKAY
```

```
<inner-paragraph> ::= <inner-text> <inner-paragraph>
                    | ε

<inner-text> ::= <variable-use>
                 | <comment>
                 | <bold>
                 | <list>
                 | <link>
                 | <newline>
                 | TEXT

<bold> ::= GIMMEH BOLD TEXT OIC

<list> ::= MAEK LIST <list-item> <list-items> MKAY

<list-items> ::= <list-item> <list-items>
                 | ε

<list-item> ::= GIMMEH ITEM <optional-variable-define> <inner-list> OIC

<inner-list> ::= <list-content> <inner-list>
                 | ε

<list-content> ::= <bold>
                   | <comment>
                   | <link>
                   | <newline>
                   | <variable-use>
                   | TEXT

<link> ::= GIMMEH LINX ADDRESS OIC

<newline> ::= NEWLINE
```

```
<optional-variable-define> ::= <variable-define>
                             | ε

<variable-define> ::= IHAZ VARNAME ITIZ VARVALUE MKAY

<variable-use> ::= GIMMEH VAR VARNAME OIC
```